

EHRMAN CREST CIRC | GRADES 5 AND 6

CIRC Class Cards

All 31 contacts. Put these on the board.

One card per step. Big text. Read it, do it, show it.

Neuro art and book checkout

Today you can join curved lines, round their corners, and follow our book-checkout routine.

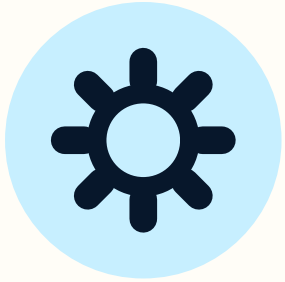
What you need. Watercolor paper ■ Black drawing markers ■ Washable color markers ■
Class artwork trays ■ Existing library books and checkout system



See today's plan

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

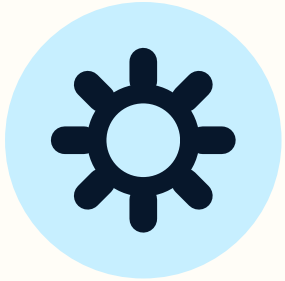
Eyes up. Here is the job.



Watch the line

- Watch four curved lines cross.
- Turn one sharp corner into a smooth curve.
- Notice the large spaces left for color.

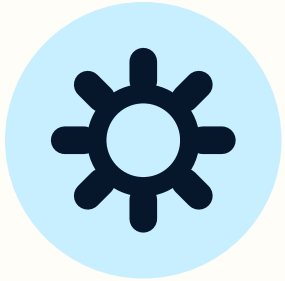
Make it. Try it. Fix one thing.



Draw and round

- Write your name and class on the back.
- Draw four to six crossing curves.
- Round the sharp corners. Add dry color if time allows.

Make it. Try it. Fix one thing.



Choose your book

- Cap markers and place artwork in the class tray.
- When called, follow our library checkout routine.
- Return with your book and read quietly.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Point to one corner you rounded and show that your table is ready.

Show me what you can do.

Turn artwork into a digital image

Today you can capture a clear image, name it clearly, and check that another person can find it.

Remember. Use the device Mr. Macek shows you. No device? Draw a paper preview and write its file label.

What you need. Existing school-approved camera or scanner ■ Existing school device and approved storage ■ Artwork ■ Paper ■ Pencils



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

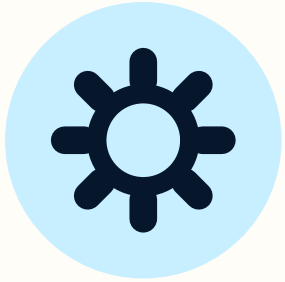
Eyes up. Here is the job.



Watch and try

- Watch a page being placed flat in good light.
- Compare a crooked image with a cropped, readable image.
- Watch the teacher save one example.

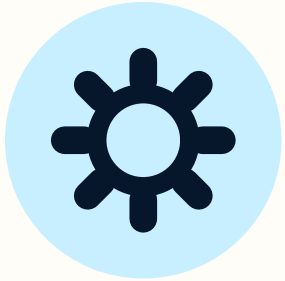
Safety first, then we build.



Make it work

- Use the school-approved camera or scanner shown by your teacher.
- Capture only your artwork. Keep faces and private information outside the image.
- Save using the class code and project label shown on the board.

Make it. Try it. Fix one thing.



Check one thing

- Find your saved image again.
- Check the edges, direction, and readability.
- If no device is available, draw a paper preview and write its file label.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show a readable saved image, or a paper preview with a clear file label.

Show me what you can do.

Design a logo people can read

Today you can combine a simple symbol and readable lettering, then improve them after a quick test.

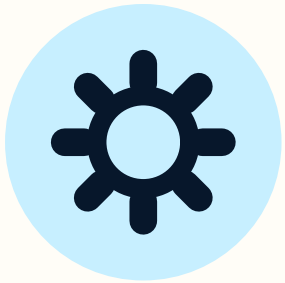
What you need. Paper ■ Pencils ■ Erasers ■ Rulers ■ Shared markers ■ Existing approved digital image tool if available



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

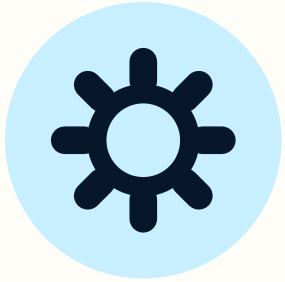
Eyes up. Here is the job.



Watch and try

- Compare a clear symbol with a crowded symbol.
- Look at the same logo large and small.
- Notice how light and dark areas help the words stand out.

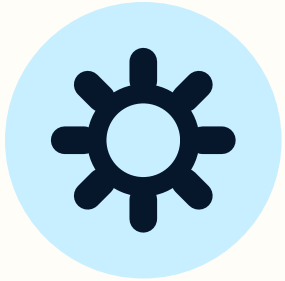
Make it. Try it. Fix one thing.



Make it work

- Choose a class project or invented product.
- Sketch three small logo ideas.
- Choose one. Use a simple symbol and no more than three colors.

Make it. Try it. Fix one thing.



Check one thing

- Show the logo without explaining it.
- Ask what the viewer notices and reads first.
- Change one unclear detail and test again.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show your logo and name the change that made it clearer.

Show me what you can do.

Cardboard joins and careful cutting

Today you can make a stable cardboard connection and test it before adding decoration.

Remember. Scissors only. Mr. Macek makes any cut that needs a blade.

What you need. Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

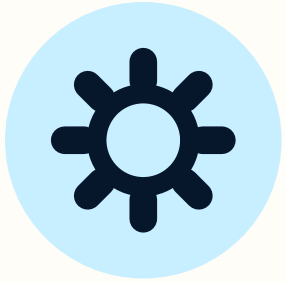
Eyes up. Here is the job.



Watch and try

- Watch the teacher cut away from the holding hand.
- Compare a tab, slot, and taped join.
- Watch the teacher test each join gently.

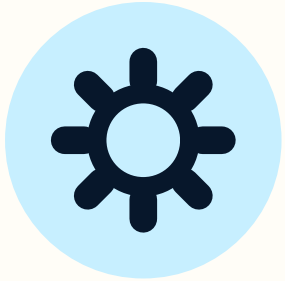
Safety first, then we build.



Make it work

- Sketch a small freestanding shape.
 - Choose a tab, slot, or taped join.
 - Cut with approved scissors or use teacher-prepared pieces.
- Assemble your shape.

Make it. Try it. Fix one thing.



Check one thing

- Place your shape on the table and test its balance.
- Change one weak join.
- Compare the first and improved versions.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show one stable connection and explain why it holds.

Show me what you can do.

Make a package that protects

Today you can protect a model product, test the package, and improve one weak point.

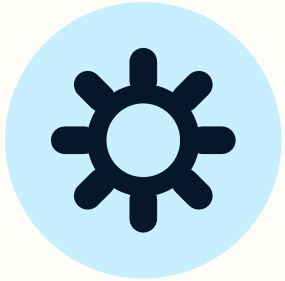
What you need. Reclaimed small boxes ■ Clean cardboard ■ Paper ■ String ■ Masking tape ■ Model products



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

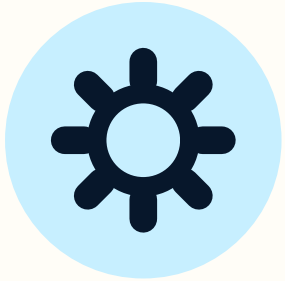
Eyes up. Here is the job.



Watch and try

- Watch a model product move inside a loose package.
- Notice padding, support, and an opening.
- Watch one teacher-approved handling test.

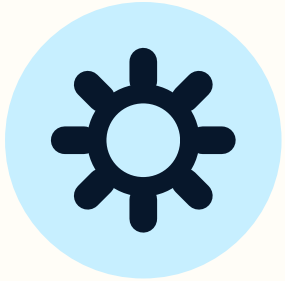
Make it. Try it. Fix one thing.



Make it work

- Choose the reusable model product provided.
- Sketch a package that opens without damage.
- Build using a material limit shown by your teacher.

Make it. Try it. Fix one thing.



Check one thing

- Run the same gentle handling test twice.
- Check movement, damage, and ease of opening.
- Improve one weak point without adding unnecessary material.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Demonstrate how your package protects and opens, and explain one improvement.

Show me what you can do.

Plan a small casting without pouring

Today you can draw a useful small object, estimate its size, and compare possible materials.

Remember. Today is planning on paper. Nobody mixes, pours, heats, or touches resin.

What you need. Paper ■ Pencils ■ Rulers ■ Calculators if available ■ Empty dry mold or dimensioned drawing ■ Teacher-cleared cured example or picture



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

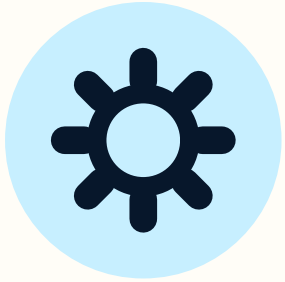
Eyes up. Here is the job.



Watch and try

- Look at a dry mold or its measured drawing.
- Identify the inside space that shapes the object.
- Compare a cardboard model with a teacher-cleared cured example or picture.

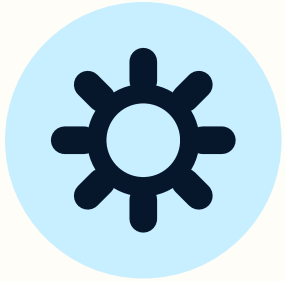
Safety first, then we build.



Make it work

- Sketch a small token, tag, or game piece.
- Measure a rectangular practice cavity or its drawing.
- Use length, width, and depth to estimate its volume.

Make it. Try it. Fix one thing.



Check one thing

- Compare making your design from cardboard and casting material.
- Name one reason for each choice.
- Improve the design using the teacher's size limit.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show a labeled design and explain why your chosen material fits its use.

Show me what you can do.

Tank 1: Notice a useful problem

Today you can notice a real, manageable problem and explain who it affects.

Remember. Pick a task or an object that could work better. Never a person.

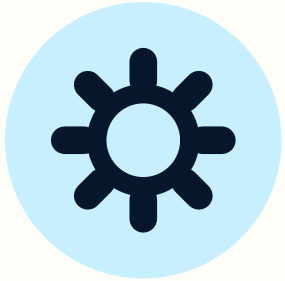
What you need. Paper ■ Pencils ■ Teacher-approved problem pictures



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

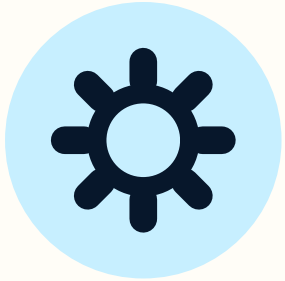
Eyes up. Here is the job.



Watch and try

- Compare a broad wish with one specific everyday problem.
- Notice who is affected and when it happens.

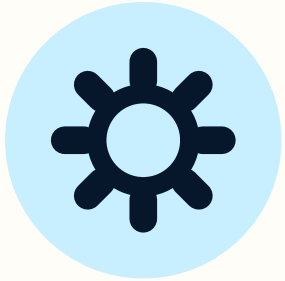
Make it. Try it. Fix one thing.



Make it work

- Look at the approved classroom examples.
- List or sketch three small problems.
- Choose one that can be explored with classroom materials.

Make it. Try it. Fix one thing.



Check one thing

- Tell a partner who has the problem.
- Explain when it happens.
- Narrow any idea that needs money, machinery, or changes outside our control.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Complete: Someone needs help with this problem when this happens.

Show me what you can do.

Tank 2: Listen before choosing

Today you can ask a useful question and use the answer to improve my problem statement.

Remember. Ask a partner or use a user card. We do not contact anyone outside this room.

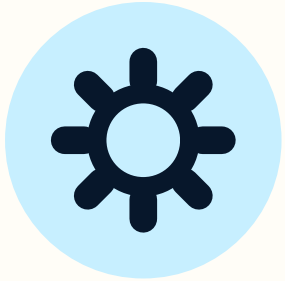
What you need. Paper ■ Pencils ■ Reusable user cards



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

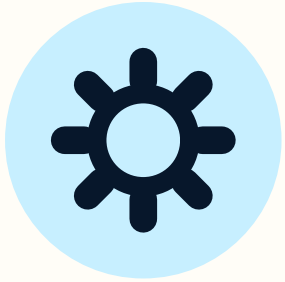
Eyes up. Here is the job.



Watch and try

- Compare a leading question with an open question.
- Watch a short respectful interview.

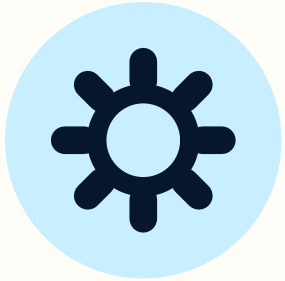
Make it. Try it. Fix one thing.



Make it work

- Prepare two questions about the task or object.
- Ask an approved partner, or use the teacher's user card.
- Record the useful answer with a short note or drawing.

Make it. Try it. Fix one thing.



Check one thing

- Compare the answer with your first idea.
- Change the problem statement if needed.
- Ask whether you are describing a need or jumping to a solution.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Share one thing you learned from the user and what it changed.

Show me what you can do.

Tank 3: Try three ideas

Today you can sketch different solutions and choose one using a clear reason.

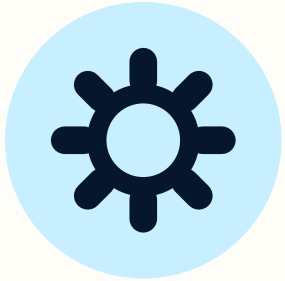
What you need. Paper ■ Pencils ■ Rulers ■ Reusable criteria cards



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

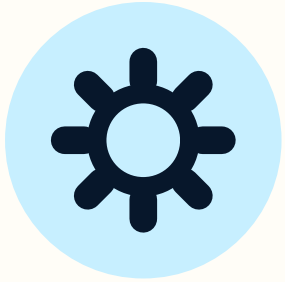
Eyes up. Here is the job.



Watch and try

- See three different ways to solve one small problem.
- Compare usefulness, simplicity, and available materials.

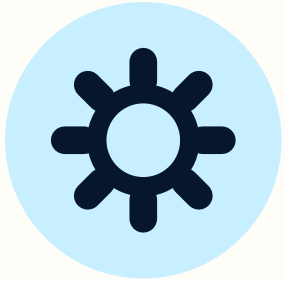
Make it. Try it. Fix one thing.



Make it work

- Sketch three different solutions.
- Label the part that does the work.
- Circle one idea you can test with classroom materials.

Make it. Try it. Fix one thing.



Check one thing

- Show the ideas to a partner.
- Name one strength and one concern for the chosen idea.
- Revise the sketch before building.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show three ideas and explain why one is the best first test.

Show me what you can do.

Tank 4: Plan the first build

Today you can make a labeled plan and choose a small, testable first version.

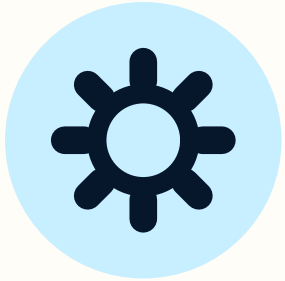
What you need. Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

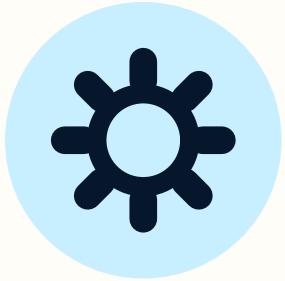
Eyes up. Here is the job.



Watch and try

- Watch a sketch become a parts list.
- Separate essential working parts from decoration.

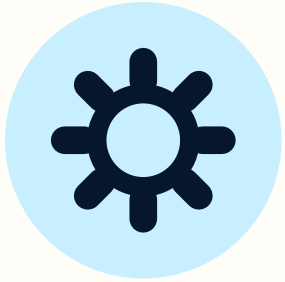
Make it. Try it. Fix one thing.



Make it work

- Draw the first version from a useful view.
- Label its working parts and approximate sizes.
- Choose materials and agree on the first build jobs.

Make it. Try it. Fix one thing.



Check one thing

- Explain how a user will try the model.
- Remove any part that does not help the first test.
- Show the teacher your plan before collecting supplies.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show a labeled plan, short materials list, and one thing your model must do.

Show me what you can do.

Tank 5: Practice the tricky part

Today you can practice one connection or moving part before using it in my prototype.

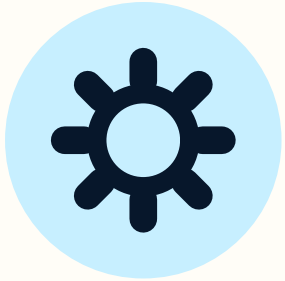
What you need. Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

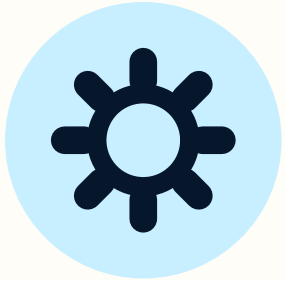
Eyes up. Here is the job.



Watch and try

- Watch a tab, hinge, slider, or string pull.
- Identify where friction, looseness, or weakness could occur.

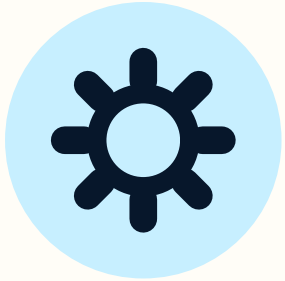
Make it. Try it. Fix one thing.



Make it work

- Choose the hardest working part in your plan.
- Build a small practice sample using scrap.
- Try the movement or connection several times.

Make it. Try it. Fix one thing.



Check one thing

- Name the first weak or unreliable point.
- Change one feature.
- Keep the stronger sample as a reference.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Demonstrate your practice part and name the improvement.

Show me what you can do.

Tank 6: Build version one

Today you can build a rough model that shows my idea's main job.

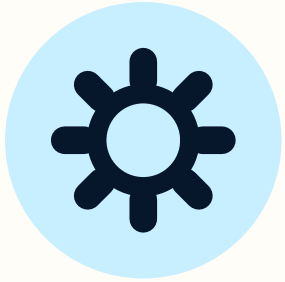
What you need. Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

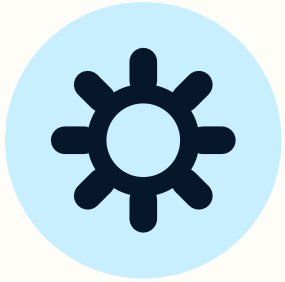
Eyes up. Here is the job.



Watch and try

- Watch the main working part being built first.
- Notice that a rough prototype can answer a useful question.

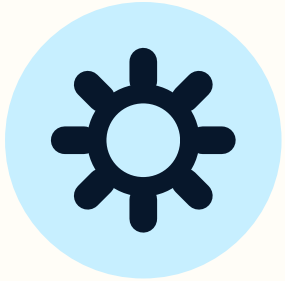
Make it. Try it. Fix one thing.



Make it work

- Build the essential working part from your plan.
- Add only the support needed to try it.
- Keep a sketch or picture of version one using the approved classroom method.

Make it. Try it. Fix one thing.



Check one thing

- Ask a partner to identify the model's job.
- Try the intended action once.
- Mark one feature that needs a test next time.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show version one doing its main job, or identify the exact part that needs repair.

Show me what you can do.

Tank 7: Plan a fair test

Today you can plan a repeatable test that checks what my design claims.

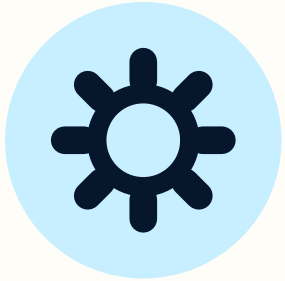
What you need. Prototype ■ Paper ■ Pencils ■ Ruler or timer if needed ■ Reusable test objects



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

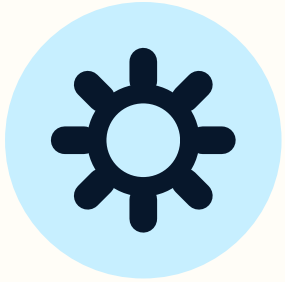
Eyes up. Here is the job.



Watch and try

- Compare two tests that change several things.
- Watch a fair test keep the object, starting point, and action consistent.

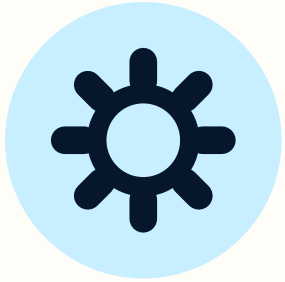
Make it. Try it. Fix one thing.



Make it work

- Write or draw what the product claims to do.
- Choose one thing to count, measure, or observe.
- Plan three trials with the same starting conditions.

Make it. Try it. Fix one thing.



Check one thing

- Walk through the test without rushing.
- Ask a partner what must stay the same.
- Get the teacher's approval before the real trials.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Explain your test, what stays the same, and what result would count as success.

Show me what you can do.

Tank 8: Test and record

Today you can run the same test three times and record what actually happens.

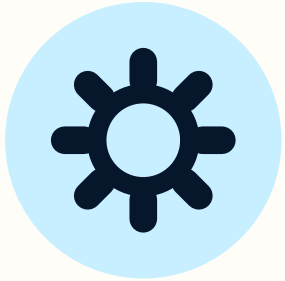
What you need. Prototype ■ Approved test objects ■ Paper ■ Pencils ■ Ruler or timer if needed



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

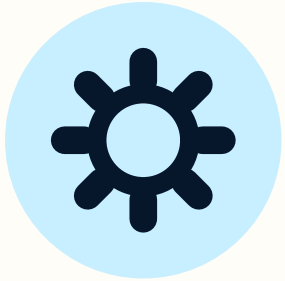
Eyes up. Here is the job.



Watch and try

- Watch one trial from the same starting position.
- See a record include an unsuccessful result honestly.

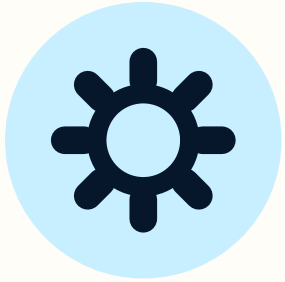
Make it. Try it. Fix one thing.



Make it work

- Reset the test to the agreed starting position.
- Run three trials and record each result.
- Keep the design unchanged until the trials are finished.

Make it. Try it. Fix one thing.



Check one thing

- Compare all three results.
- Point to the most useful pattern or failure.
- Choose one problem to fix next.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show three results and identify one specific design problem.

Show me what you can do.

Tank 9: Improve one feature

Today you can use test evidence to change one feature and check whether it helps.

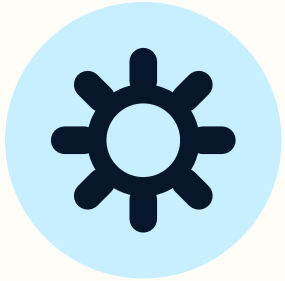
What you need. Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

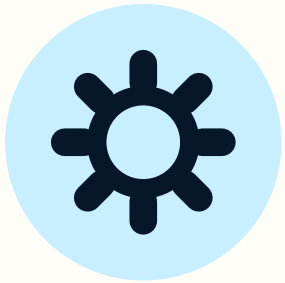
Eyes up. Here is the job.



Watch and try

- Compare a design change with a decorative change.
- Trace one failed test to the part causing it.

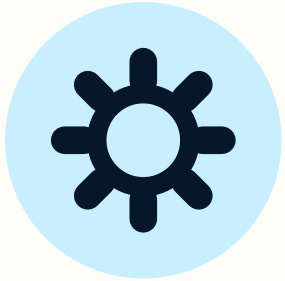
Make it. Try it. Fix one thing.



Make it work

- Choose one problem from your test record.
- Change the relevant shape, connection, or movement.
- Repeat the same three-trial test.

Make it. Try it. Fix one thing.



Check one thing

- Compare the old and new results.
- Keep the improvement or explain why you will undo it.
- Record what the evidence suggests next.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show one evidence-based change and compare its before-and-after results.

Show me what you can do.

Tank 10: Let someone try

Today you can watch a user try my idea and learn without coaching every move.

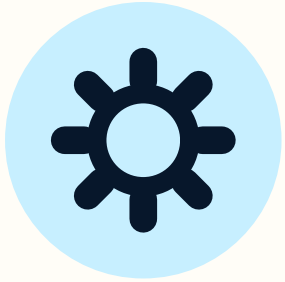
What you need. Prototype ■ Paper ■ Pencils ■ Reusable feedback cards ■ Scrap materials



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

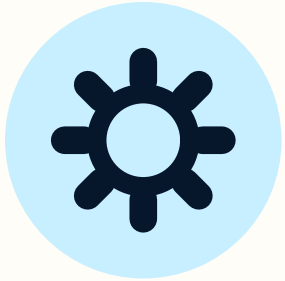
Eyes up. Here is the job.



Watch and try

- Watch a user receive one short instruction.
- Notice how the designer listens without taking over.

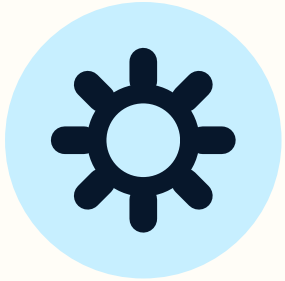
Make it. Try it. Fix one thing.



Make it work

- Tell an approved partner the product's job.
- Let the partner try it using the approved test.
- Record one clear action, one confusion, and one useful comment.

Make it. Try it. Fix one thing.



Check one thing

- Ask what was easiest and hardest.
- Choose one change that would help the user.
- Try a quick paper or scrap version of that change.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Explain one user difficulty and the change you will make.

Show me what you can do.

Tank 11: Use materials wisely

Today you can choose materials that do the job while reducing unnecessary waste.

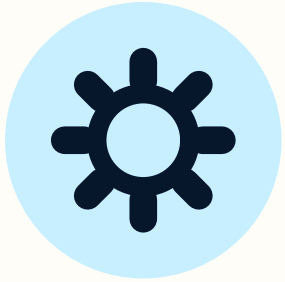
What you need. Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

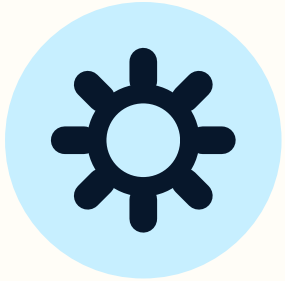
Eyes up. Here is the job.



Watch and try

- Compare two materials doing the same small job.
- Notice what can be reused or made smaller.

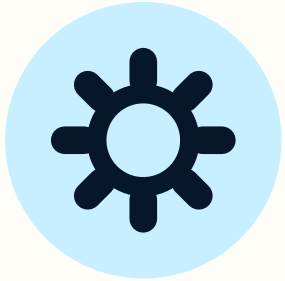
Make it. Try it. Fix one thing.



Make it work

- Sort the design's parts into essential and optional.
- Compare two suitable materials for one part.
- Improve that part without making the product harder to use.

Make it. Try it. Fix one thing.



Check one thing

- Check that the main job still works.
- Count pieces or compare material size before and after.
- Sort useful scraps for another group.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show a useful material choice and explain what you saved or improved.

Show me what you can do.

Tank 12: Name and package

Today you can communicate my product's purpose with a clear name, logo, and simple package.

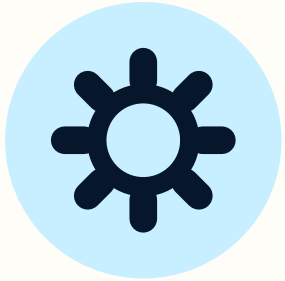
What you need. Prototype ■ Paper ■ Shared markers ■ Reclaimed boxes ■ Cardboard ■ Tape



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

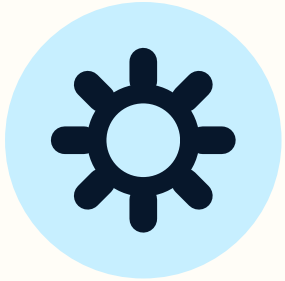
Eyes up. Here is the job.



Watch and try

- Compare a readable product label with a crowded one.
- Watch a package protect the model and open easily.

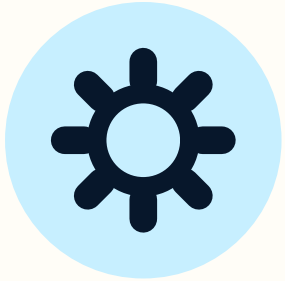
Make it. Try it. Fix one thing.



Make it work

- Choose a clear product name and reuse your logo skills.
- Make a label that tells the product's job.
- Create or improve a simple protective package.

Make it. Try it. Fix one thing.



Check one thing

- Ask a partner to read the label without help.
- Check opening, fit, and the approved handling test.
- Revise one confusing or wasteful detail.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Show a readable product label and a package that opens and protects.

Show me what you can do.

Tank 13: Explain your idea

Today you can explain the problem, my solution, and evidence that helped me improve it.

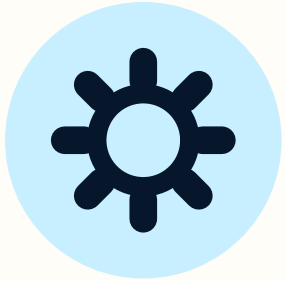
What you need. Prototype ■ Previous test record ■ Index cards ■ Pencils ■ Product label



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

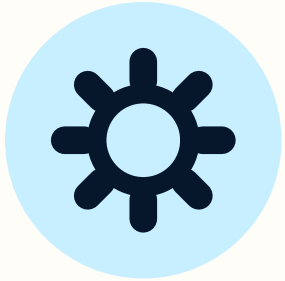
Eyes up. Here is the job.



Watch and try

- Hear a short pitch with a problem, idea, proof, and next step.
- Notice a speaker showing the product while explaining it.

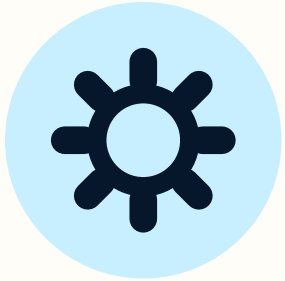
Make it. Try it. Fix one thing.



Make it work

- Make four short cue cards or picture prompts.
- Choose one clear test result as proof.
- Prepare a simple demonstration of the product.

Make it. Try it. Fix one thing.



Check one thing

- Tell the story to one partner.
- Ask the partner to repeat the main idea.
- Cut details that hide the problem, solution, or proof.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Give a short practice pitch that includes the problem, solution, and one piece of evidence.

Show me what you can do.

Tank 14: Rehearse and refine

Today you can practice a clear presentation and improve it after specific feedback.

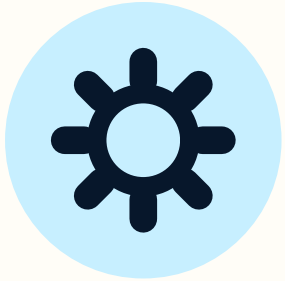
What you need. Prototype ■ Cue cards ■ Existing classroom timer ■ Reusable feedback cards ■ Display surface



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

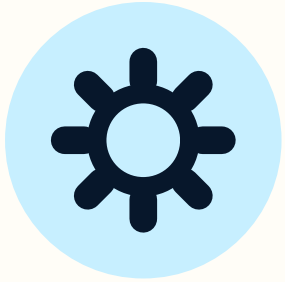
Eyes up. Here is the job.



Watch and try

- Watch a calm handoff and a table-level demonstration.
- Compare useful feedback with a vague compliment.

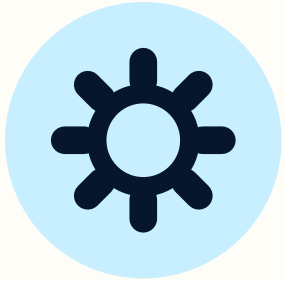
Make it. Try it. Fix one thing.



Make it work

- Choose who will explain or demonstrate each part.
- Rehearse once while a partner times the explanation.
- Practice one likely question and an honest answer.

Make it. Try it. Fix one thing.



Check one thing

- Ask for one strength and one specific improvement.
- Change the unclear part.
- Rehearse again with a reliable reset.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Deliver your revised pitch and show how the product resets for the next visitor.

Show me what you can do.

Tank 15: Share and reflect

Today you can show my design, listen to feedback, and explain how it changed.

Remember. This is an in-class share. You present to your classmates.

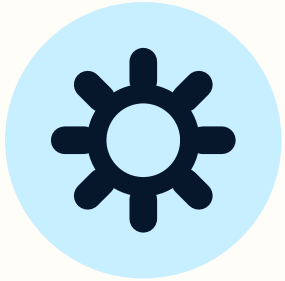
What you need. Prototype ■ First sketch ■ Test evidence ■ Labels ■ Existing display surfaces ■ Paper



Get ready

- Read today's goal.
- Look at the example.
- Tell your partner one useful detail.

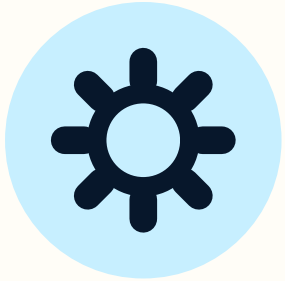
Eyes up. Here is the job.



Watch and try

- Watch a visitor approach, hear a short pitch, and ask one question.
- Notice how the presenter resets the space.

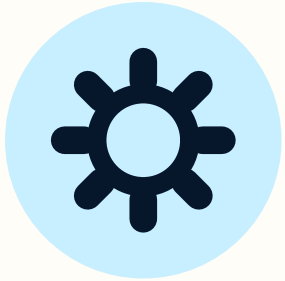
Make it. Try it. Fix one thing.



Make it work

- Set out the prototype, label, and one piece of test evidence.
- Take turns presenting and visiting other teams.
- Give one useful observation and one respectful question.

Make it. Try it. Fix one thing.



Check one thing

- Compare your first sketch with your final version.
- Choose the most useful change.
- Name one thing you would test next.

Make it. Try it. Fix one thing.



Reset together

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

Leave it better than you found it.



Show your result

- Explain how evidence changed your design and identify a sensible next test.

Show me what you can do.

Outdoor Microclimate Map

Today: Observe and compare outdoor places. Thermometers are optional.

Remember. Thermometers are optional. If we do not have one, record what you can see and feel.

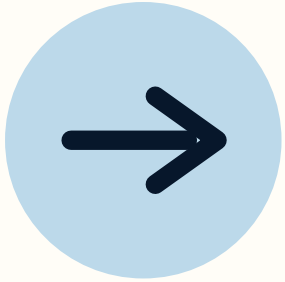
What you need. Clipboards ■ Outdoor zone map ■ Pencils ■ Thermometers if available ■
Wind ribbons ■ Observation key



Get Ready

- Clip the zone map to your board.
- Check your assigned zones.

Eyes up. Here is the job.



Walk Out

- Walk with your weather team.

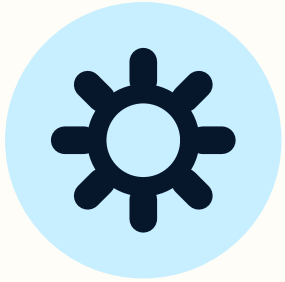
Move together. Stay with your team.



Use Tools Safely

- Carry the thermometer by its case.
- Leave unknown plants untouched.
- Stay inside each marked zone.

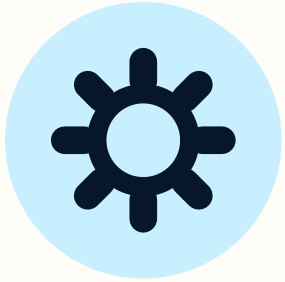
Safety first, then we build.



Observe Two Zones

- Mark sun or shade.
- Mark the surface condition.
- Watch the wind ribbon.

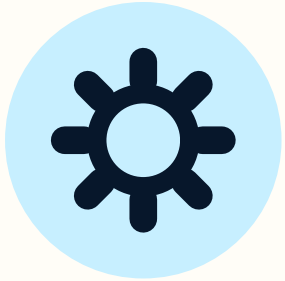
Make it. Try it. Fix one thing.



Take Readings

- Take one approved reading in each zone.
- Write each number on the map.

Make it. Try it. Fix one thing.



Build The Map

- Add both zones to the class key.
- Choose the best zone for one activity.

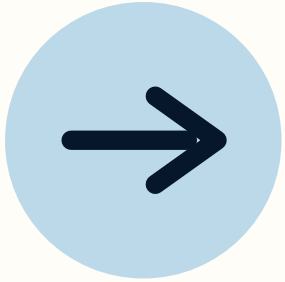
Make it. Try it. Fix one thing.



Pack The Tools

- Return each tool to its case.
- Clip all maps together.

Leave it better than you found it.



Return Inside

- Walk back with your weather team.

Move together. Stay with your team.



Exit Check

- Write two observations that support your zone choice.

Show me what you can do.



Get Ready

- Place the indoor map on your table.
- Pick up two weather data cards.

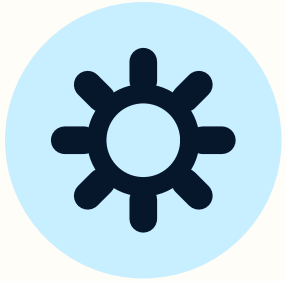
Eyes up. Here is the job.



Keep Cards Safe

- Keep every card on the table.
- Use no outdoor tool.
- Walk only when the teacher says go.

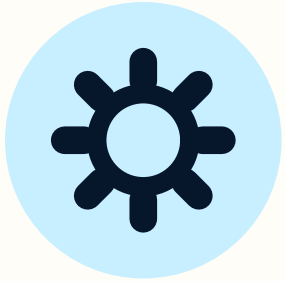
Safety first, then we build.



Learn The Map Key

- Find the sun mark.
- Find the shade mark.
- Find the wind mark.

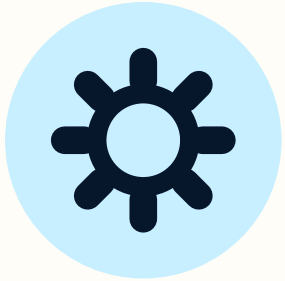
Make it. Try it. Fix one thing.



Compare Two Zones

- Read the first weather data card.
- Mark its sun, wind, and ground clues.
- Do the same with the second card.

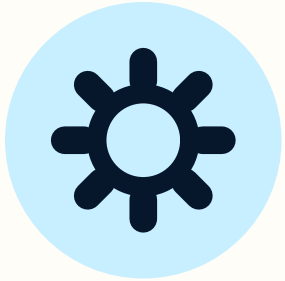
Make it. Try it. Fix one thing.



Build The Map

- Add both zones to the indoor map.
- Copy each number carefully.
- Check the map with a partner.

Make it. Try it. Fix one thing.



Choose A Zone

- Pick the best zone for one activity.
- Circle two clues that support your choice.

Make it. Try it. Fix one thing.



Reset The Table

- Stack the weather data cards.
- Return the map and pencil.

Leave it better than you found it.



Exit Check

- Name the zone you picked.
- Read your two clues.

Show me what you can do.

Seed Travelers

Today: Test a seed-inspired carrier with shared test stations.

What you need. Teacher-provided seed samples or image cards ■ Paper ■ Coffee filters ■ Cotton ■ Yarn ■ Paper clips



Get Ready

- Choose one seed sample card.
- Place the test sheet on your table.

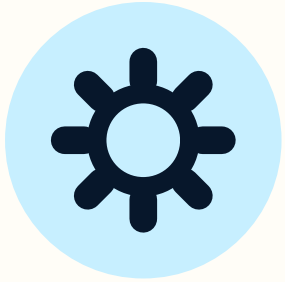
Eyes up. Here is the job.



Test Safely

- Keep water inside the tray.
- Wipe each spill with a towel.
- Keep every sample away from your mouth.

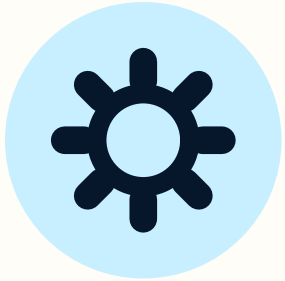
Safety first, then we build.



Sort The Seeds

- Study the seed shape.
- Choose wind, drop, float, or hitchhiking.
- Record your reason.

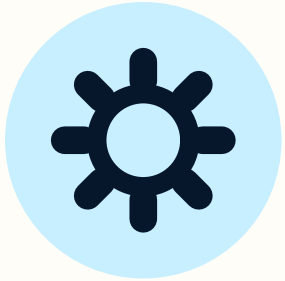
Make it. Try it. Fix one thing.



Build A Model

- Sketch one model seed.
- Choose materials that match the travel method.
- Build the model.

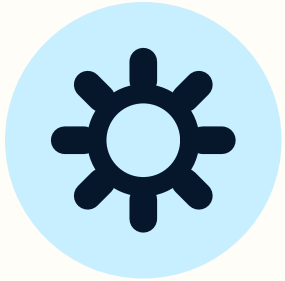
Make it. Try it. Fix one thing.



Run Three Trials

- Use the matching test station.
- Run one trial at a time.
- Record each result.

Make it. Try it. Fix one thing.



Change One Part

- Change one shape or material.
- Repeat the same test.

Make it. Try it. Fix one thing.



Reset The Station

- Dry the wet station.
- Return every model piece.

Leave it better than you found it.



Exit Check

- Circle the better result.
- Name the useful feature.

Show me what you can do.

Paper Bridge

Today: Compare a small structure using the same reusable loads.

What you need. Copy paper ■ Index cards ■ Tape ■ Two support blocks ■ Uniform load pieces ■ Recording sheet



Get Ready

- Place the support blocks on the marks.
- Count the load pieces.

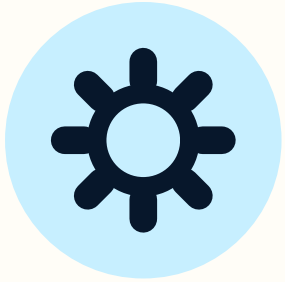
Eyes up. Here is the job.



Test Safely

- Keep hands below the table edge.
- Place only approved loads on the bridge.

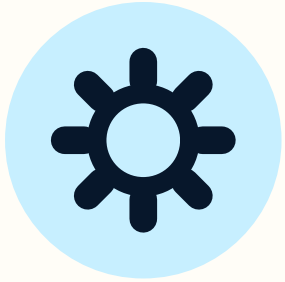
Safety first, then we build.



Plan The Span

- Choose a beam shape.
- Predict where the paper may bend.
- Sketch the bridge.

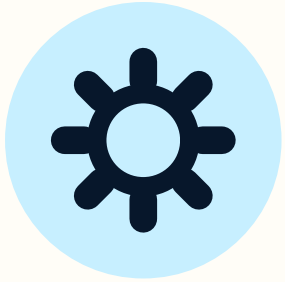
Make it. Try it. Fix one thing.



Build The Bridge

- Fold the paper into your chosen shape.
- Tape only the planned joints.
- Place the bridge across the fixed span.

Make it. Try it. Fix one thing.



Load The Bridge

- Add one load piece.
- Record the supported total.
- Repeat until the bridge fails.
- Change one weak part.

Make it. Try it. Fix one thing.



Reset The Test

- Return every load piece.
- Recycle damaged paper.

Leave it better than you found it.



Exit Check

- Write the highest supported load.
- Name your useful change.

Show me what you can do.

Moving Picture Machine

Today: Make and revise a short paper animation.

What you need. Index cards ■ Paper strips ■ Cardstock circles ■ Pencils ■ Markers ■
Brass fasteners



Get Ready

- Choose flip, wheel, or strip motion.
- Place the frame paper in order.

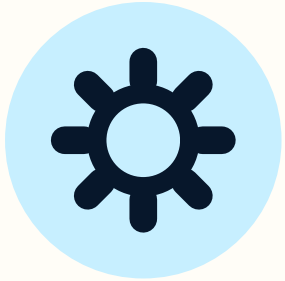
Eyes up. Here is the job.



Fastener Safety

- Keep the device at table level.
- Use fasteners only through marked holes.
- Ask the teacher about a sharp fastener.

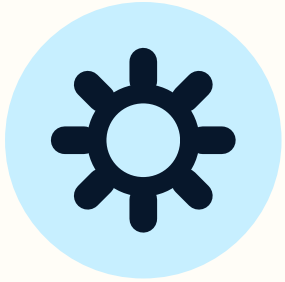
Safety first, then we build.



Plan The Motion

- Draw a clear start pose.
- Draw a clear finish pose.
- Mark the matching dot.

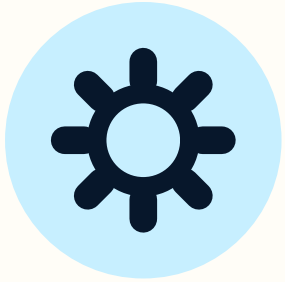
Make it. Try it. Fix one thing.



Draw The Frames

- Change one small part per frame.
- Keep each drawing on the registration mark.
- Number the frames.

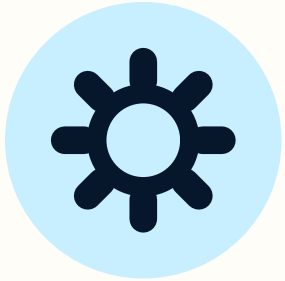
Make it. Try it. Fix one thing.



Assemble The Motion

- Stack or attach the frames in order.
- Secure the approved fastener.
- Move the device slowly.

Make it. Try it. Fix one thing.



Smooth The Motion

- Find the largest visual jump.
- Revise one frame.
- Test at two speeds.

Make it. Try it. Fix one thing.



Sort The Pieces

- Return unused fasteners.
- Clip your frames together.

Leave it better than you found it.



Exit Check

- Show the finished motion once.
- Name the frame you changed.

Show me what you can do.

Hydroponics Flow Lab

Today: Test a gravity-fed water model, not a purchased growing installation.

What you need. Clean tubing ■ Cups ■ Reclaimed channels ■ Water trays ■ Measuring cups ■ Towels



Get Ready

- Place the dry tray on the wet mat.
- Measure the test water.

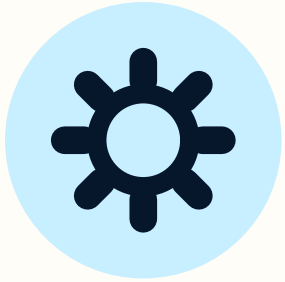
Eyes up. Here is the job.



Wet Tool Safety

- Keep water inside the tray.
- Wipe each spill with a towel.
- Do not cut tubing yourself.
- Keep electronics away from the wet zone.

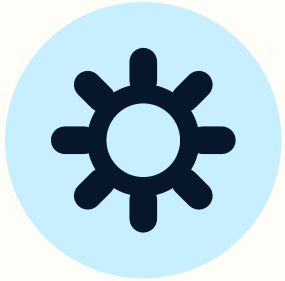
Safety first, then we build.



Plan The Flow

- Draw the water cup.
- Draw the water path.
- Mark each plant spot and the end cup.

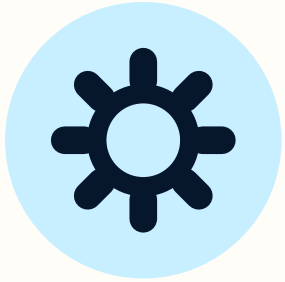
Make it. Try it. Fix one thing.



Build The Dry Model

- Mark the cut line on the tube.
- Give the tube to the teacher.
- Wait for the cut, then connect the tube path.
- Check every connection while dry.

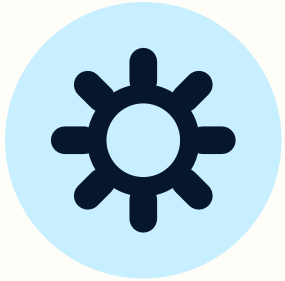
Make it. Try it. Fix one thing.



Run The Water

- Pour the measured water into the water cup.
- Watch each plant spot.
- Measure the collected water.

Make it. Try it. Fix one thing.



Change The Flow

- Change one slope, hole, or loose connection.
- Repeat with the same water amount.
- Record the new collected amount.

Make it. Try it. Fix one thing.



Dry The Lab

- Drain the tray into the sink bucket.
- Place wet parts on the drying tray.
- Dry the table and floor.

Leave it better than you found it.



Exit Check

- Write the collected amount.
- Name the flow change.

Show me what you can do.

Stormwater Rescue

Today: Use shared trays to compare runoff before and after one change.

What you need. Sloped trays ■ Model soil or sand ■ Measured water cups ■ Collection cups ■ Gravel ■ Sponge



Get Ready

- Place the sloped tray on the wet mat.
- Measure the rain water.

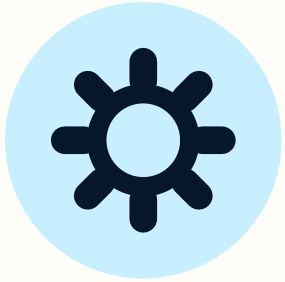
Eyes up. Here is the job.



Stormwater Safety

- Keep wet material inside the tray.
- Wipe each spill with a towel.
- Cut mesh only on the table.
- Keep fingers outside the cutting path.
- Taste no test material.

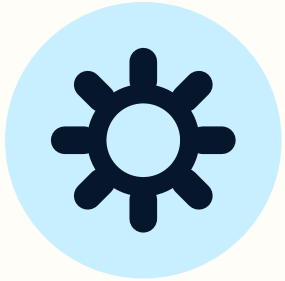
Safety first, then we build.



Run The First Test

- Pour the measured rain from the marked height.
- Measure the water that leaves the tray.
- Mark where soil moved.

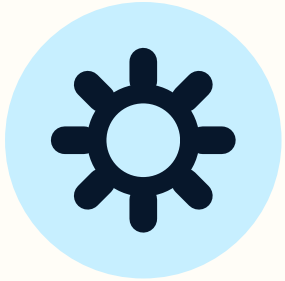
Make it. Try it. Fix one thing.



Build A Rescue

- Choose one problem location.
- Cut only the planned mesh piece.
- Build one helper that slows or moves the water.

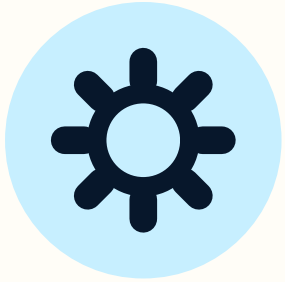
Make it. Try it. Fix one thing.



Repeat The Rain

- Pour the same water amount.
- Measure the water that leaves the tray.
- Mark where soil moved this time.

Make it. Try it. Fix one thing.



Improve The Rescue

- Change one part of the helper.
- Predict the next result.

Make it. Try it. Fix one thing.



Clean The Tray

- One table at a time, carry wet soil to the return tub.
- Place wet tools on the drying mat.
- Dry the tray and table.
- Wash hands when the teacher calls your table.

Leave it better than you found it.



Exit Check

- Compare both water amounts.
- Name the helpful feature.

Show me what you can do.

Micro:bit Sensor Station

Today: Use existing approved devices or the complete paper-code fallback.

Remember. No device today? The paper-code version teaches the same idea and counts the same.

What you need. Classroom micro:bits or confirmed equivalents ■ USB cables ■ Computers ■ Teacher-approved sensors if available ■ Starter code cards ■ Sensor data cards



Get Ready

- Place the micro:bit beside the computer.
- Open the teacher starter file.
- Connect only the approved cable.
- Wait until the computer shows the board.

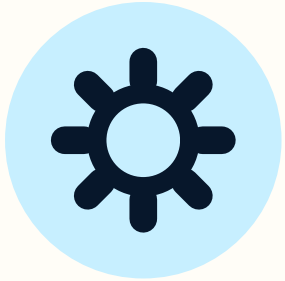
Eyes up. Here is the job.



Device Safety

- Use only approved USB power.
- Connect no unknown device.
- Handle the board by its edges.
- Disconnect power when the teacher says stop.

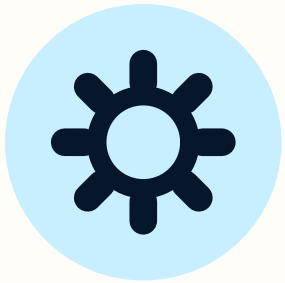
Safety first, then we build.



Plan The Sensor

- Choose one classroom condition.
- Write what the board will measure.
- Choose the number that turns the alert on.
- Choose a clear alert.

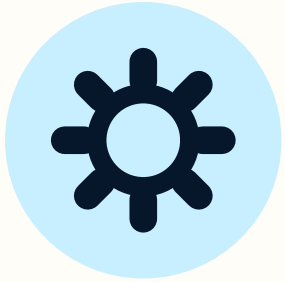
Make it. Try it. Fix one thing.



Build The Program

- Place the measure block first.
- Add the number check.
- Add the alert block.
- Transfer the program to the approved device.

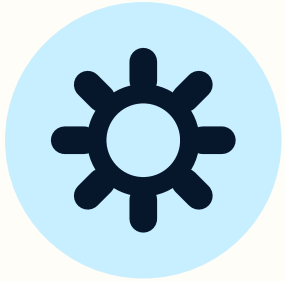
Make it. Try it. Fix one thing.



Test Three Values

- Test one number below the turn-on number.
- Test the turn-on number.
- Test one number above it.
- Record each alert.

Make it. Try it. Fix one thing.



Tune The Alert

- Change the turn-on number or alert.
- Repeat the three tests.

Make it. Try it. Fix one thing.



Disconnect Safely

- Stop the program.
- Unplug the USB cable by its connector.
- Return the board and cable to the tray.

Leave it better than you found it.



Exit Check

- State the final turn-on number.
- Name one limit of the data.

Show me what you can do.



Get Ready

- Place the paper code blocks on the mat.
- Pick up three sensor data cards.

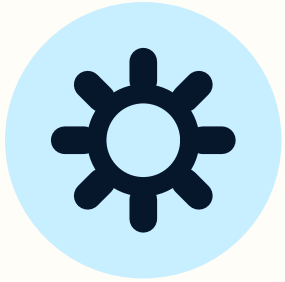
Eyes up. Here is the job.



Keep Pieces Safe

- Keep every paper block on the mat.
- Use no cable or device.
- Pick up dropped cards at once.

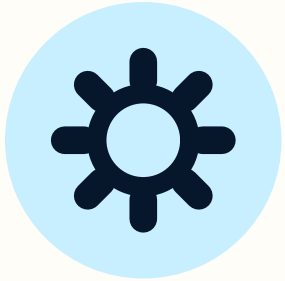
Safety first, then we build.



Plan The Alert

- Pick one classroom condition.
- Pick the number that turns the alert on.
- Draw the alert.

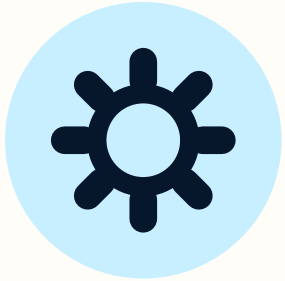
Make it. Try it. Fix one thing.



Build Paper Code

- Place the measure block first.
- Place the number check next.
- Place the alert block last.
- Read the paper code aloud.

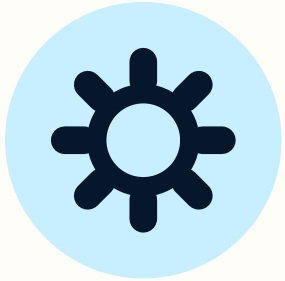
Make it. Try it. Fix one thing.



Test Three Cards

- Read the first sensor data card.
- Run it through the paper code.
- Record the alert.
- Repeat for two more cards.

Make it. Try it. Fix one thing.



Fix The Alert

- Change the turn-on number or alert.
- Test the three cards again.

Make it. Try it. Fix one thing.



Reset The Cards

- Sort the paper code blocks.
- Stack the sensor data cards.

Leave it better than you found it.



Exit Check

- Write the final turn-on number.
- Name one card that made the alert turn on.

Show me what you can do.

Cardboard Arcade

Today: Combine the year's connection and testing skills in a compact game.

What you need. Cardboard ■ Paper cups ■ Craft sticks ■ Paper fasteners ■ String ■ Tape



Get Ready

- Choose one tabletop game card.
- Place tokens inside the play tray.

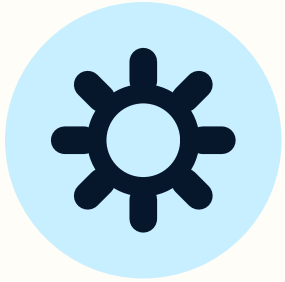
Eyes up. Here is the job.



Game Safety

- Keep tokens inside the table boundary.
- Stop any part that pinches.
- Use no launcher.

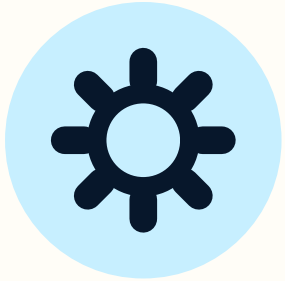
Safety first, then we build.



Plan The Game

- Write the game goal in one sentence.
- Sketch the token path.

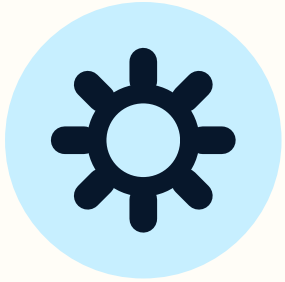
Make it. Try it. Fix one thing.



Build The Game

- Build the smallest playable version.
- Add one reliable cardboard connection.
- Make a clear start point.

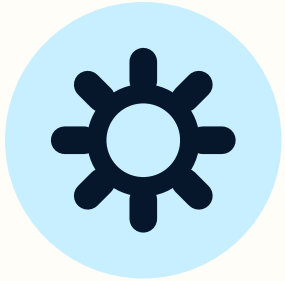
Make it. Try it. Fix one thing.



Play Test

- Ask a partner to play without help.
- Watch the first confusing moment.
- Record one unreliable part.

Make it. Try it. Fix one thing.



Improve The Reset

- Change the confusing part.
- Reset the game twice.

Make it. Try it. Fix one thing.



Reset The Arcade

- Count every token.
- Place the game on the storage shelf.

Leave it better than you found it.



Exit Check

- State the goal.
- Show how the game resets.

Show me what you can do.

Tech Terrarium

Today: Make a teacher-approved shared contribution while preserving the existing habitat.

Remember. The terrarium stays exactly as it is. Build your new part off to the side.

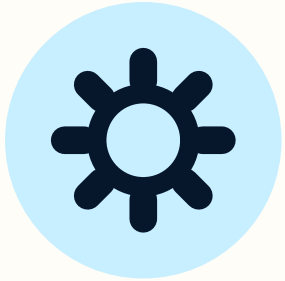
What you need. Teacher-approved collected rocks ■ Wash tubs ■ Water ■ Towels ■
Terrarium container ■ Prepared base materials



Get Ready

- Look at the saved terrarium.
- Touch nothing yet.

Eyes up. Here is the job.



Choose Jobs

- Read one refresh job card.
- Go to your team spot.

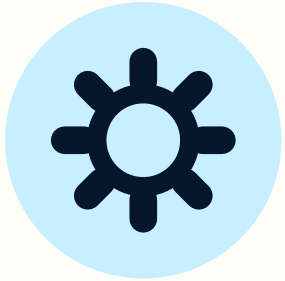
Make it. Try it. Fix one thing.



Keep It Safe

- Keep all glued parts together.
- Do not pull or pry.
- Move only loose parts the teacher marked.

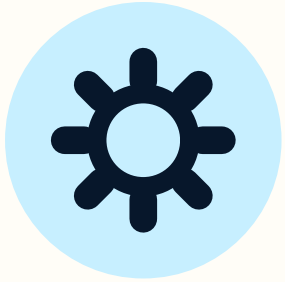
Safety first, then we build.



Look And Record

- Find one part that should stay.
- Find one spot that can work better.
- Draw what you see.

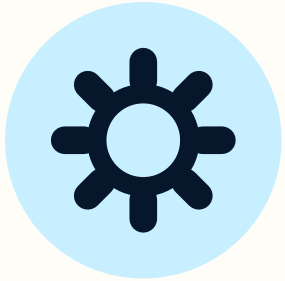
Make it. Try it. Fix one thing.



Plan One Add On

- Draw one part that can come back off.
- Keep space for plants and future tools.

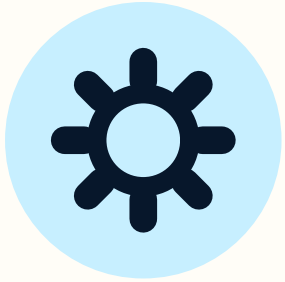
Make it. Try it. Fix one thing.



Build Off To The Side

- Build the new part away from the terrarium.
- Use only the tools on your card.
- Wait for the teacher check.

Make it. Try it. Fix one thing.



Test The New Part

- Test the new part away from the terrarium.
- Fix one loose or crowded spot.

Make it. Try it. Fix one thing.



Reset Materials

- Return every tool.
- Store the new part by itself.

Leave it better than you found it.



Exit Check

- Name one part that stayed.
- Name the new part your team made.

Show me what you can do.

Reflect, reset, and catch up

Today: Use Demo Day reflection and reset, or complete a missed core contact.

Remember. Visitor means a classmate rotating in from another station. This is our own room.

What you need. Showcase exhibits ■ Role cards ■ Safety checklist ■ Visitor prompts ■
Feedback forms ■ Reset supplies



Get Ready

- Report to your assigned station.
- Read your role card.

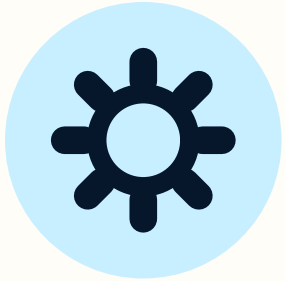
Eyes up. Here is the job.



Station Safety

- Operate only approved demonstrations.
- Keep every walkway clear.
- Pause any station that cannot reset safely.

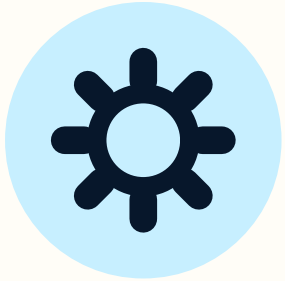
Safety first, then we build.



Check The Station

- Inspect the prototype.
- Check the visitor boundary.
- Practice the emergency pause.

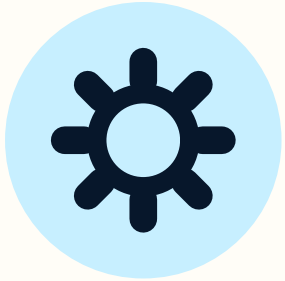
Make it. Try it. Fix one thing.



Host The Demo

- Welcome the visitor.
- State the design need.
- Demonstrate the approved action.
- Show the evidence.

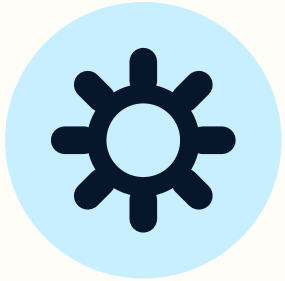
Make it. Try it. Fix one thing.



Rotate The Roles

- Reset after each visitor.
- Move to the next role card.
- Answer questions with evidence.
- Name a limit when needed.

Make it. Try it. Fix one thing.



Close The Station

- Thank the final visitor.
- Complete the last safe reset.

Make it. Try it. Fix one thing.



Restore The Room

- Return all role cards.
- Place every exhibit in storage position.

Leave it better than you found it.



Exit Check

- Name one strong visitor moment.
- Write one improvement for next time.

Show me what you can do.